

Conditional structures

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if... then... endif

It is an instruction that allows executing a **set of actions** based on the value of a **condition**. It is presented in the following form:

```
if <condition> then
```

```
    Action 1 ;
```

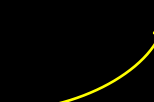
```
    Action 2 ;
```

```
    ...
```

```
    Action n ;
```

```
endif;
```

*Actions
executed if
condition= true*



<condition>: boolean
expression

If the condition is **verified (condition=true)**, the group of actions (**instruction1 to instruction n**) is executed, otherwise no instruction is executed.

Example1

- The maximum of two integers A and B.

```
Algorithm maximum;
var
    a, b , max: integer;
Begin
    write(" Give two integer values :");
    read(a ,b);
    if a <= b then
        max← b;
    endif;
    if b < a then
        max← a;
    endif;
    write(" the maximum between", a," and ", b, " is: ", max);
end.
```

Exemple1 en C

- The maximum of two integers A and B.

```
int main() {  
    printf(" Give two integer values :\n");  
    int a,b, max;  
    scanf("%d%d",&a,&b);  
    if(a<=b) {  
        max=b;  
    }  
    if(a>b) {  
        max=a;  
    }  
    printf("the maximum between %d and %d is: %d\n",  
a,b,max);  
    return 0;  
}
```

if... then ...else... endif

- It consists of executing two groups of actions depending on the value of a condition. It takes the following form:

```
if <condition> then
```

```
Action 11 ;  
Action 12 ;  
...  
Action 1n ;
```

*Actions is
executed if
condition= true*

```
else
```

```
Action 21 ;  
Action 22 ;  
...  
Action 2n ;
```

*Actions executed if
condition= false*

```
endif;
```

If the **condition is true**, the first group of actions (action11 through action1n) is executed.

otherwise (i.e. when **condition is false**), the second group of actions (action21 through action1n) is executed.

Example2

- Same as previous with **if..then..else**

```
Algorithm maximum;
var
  a, b , max: integer;
begin
  write("Give two ineger
values:");
  read(a ,b);
  if a <= b then
    max← b;
  endif;
  if b < a then
    max← a;
  endif;
  write("the maximum
between", a," and ", b, "
is: ", max);
end.
```

```
Algorithm maximum;
var
  a, b , max: integer;
begin
  write("Give two ineger
values:");
  read(a ,b);
  if a <= b then
    max← b;
  else
    max← a;
  endif;
  write("the maximum
between", a," and ", b, "
is: ", max);
end.
```

Example2 in C

```
int main() {
    printf("Give two integers
values:\n");
    int a,b, max;
    scanf("%d%d",&a,&b);
    if(a<=b) {
        max=b;
    }

    if(a>b) {
        max=a;
    }

    printf("the maximum between %d
and %d is: %d\n", a,b,max);
    return 0;
}
```

```
int main() {
    printf("Give two integers
values:\n");
    int a,b, max;
    scanf("%d%d",&a,&b);
    if(a<=b) {
        max=b;
    }

    else {
        max=a;
    }

    printf("the maximum between %d
and %d is: %d\n", a,b,max);
    return 0;
}
```

Example3

- the min between three numbers a, b and c

```
Algorithm minimum3nombre;  
var  
    a, b , c, min: integer;  
begin  
    write("Give three ineger values:");  
    read(a ,b ,c);  
    if a <= b and a <= c then  
        min← a;  
    else  
        if b <= a and b <= c then  
            min← b;  
        else  
            min← c;  
        endif;  
    endif;  
    write("the minimum between", a," ",  
b, " and ", c," is: ", min);  
end.
```

```
Algorithm minimum3nombre;  
var  
    a, b , c, min: integer;  
begin  
    write("Give three ineger values:");  
    read(a ,b ,c);  
    if a < b then  
        min← a;  
    else  
        min← b;  
    endif;  
    if c < min then  
        min← c;  
    endif;  
    write("the minimum between", a," ",  
b, " and ", c," is: ", min);  
end.
```

Exemple3 in C

```
//solution #1
#include <stdio.h>
int main(){
    printf("donner trois integer
values:\n");
    int a,b,c, min;
    scanf("%d%d%d",&a,&b,&c);
    if(a<=b) {
        min=a;
    }
    else {
        min=b;
    }
    if(c<=min) {
        min=c;
    }
    printf("the minimum between %d, %d
and %d is: %d\n",a,b,c,min);
    return 0;
}
```

```
//solution #2
#include <stdio.h>
int main(){
    printf(" Give three integer
values:\n");
    int a,b,c, min;
    scanf("%d%d%d",&a,&b,&c);
    if(a<=b && a<=c) {
        min=a;
    }
    else {
        if(b<=a && b<=c){
            min=b;
        }
        else {
            min=c;
        }
    }
    printf("the minimum between %d, %d
and %d is: %d\n",a,b,c,min);
    return 0;
}
```

case ..of

The case statement allows you to choose from several possibilities depending on the value of an expression, which must be of ordinal type (integer, character or Boolean).

```
<initialisation>  
Case <expression> of  
    domaine 1 : <bloc_instruction 1>;  
    domaine 2 : <bloc_instruction 2>;  
    ...  
    domaine n : <bloc_instruction n>;  
    else <autre_bloc_instruction>;  
end;
```

Remarque:

the **else** in **case .. of** statement is optional

- <expression> :is an evaluable expression.
- domain : is a list of constants.
- In this instruction, the expression is evaluated:
 - if the result is equal to an element in domain1 then the actions in block 1 are executed.
 - if the result is equal to an element of domain n then the actions of block n are executed.
- In this case if no element of any domain is equal to <expression> then the other_block will be executed.

Exemple4

- We want to write the algorithm that displays the names of the wilayas according to their codes.
- We consider only the codes: 16, 31, 30, 25.

Exemple4

- Solution with **if.. else endif** structure:

```
Algorithme wilaya;
var
    w: integer;
begin
    write(" Give a code: "); read(w);
    if w=16 then
        write(" Alger ");
    else
        if w=31 then
            write(" Oran ");
        else
            if w=30 then
                write(" Ouargla ");
            else
                if w=25 then
                    write(" Constantine ");
                else
                    write(" invalide code ");
                endif;
            endif;
        endif;
    endif;
end.
```

Exemple4

Solution by using **case .. of** structure:

```
Algorithme wilaya;  
var  
    w: integer;  
begin  
    write(" Give a code: " );  
    read(w);  
  
    case w of  
        16: write(" Alger ");  
        31: write(" Oran ");  
        25: write(" Constantine " );  
        30: write(" Ouargla ");  
        else write(" invalide code " );  
    end;  
end.
```

Exemple4

```
int main() {  
    int w;  
    printf(" Give a code : " );  
    scanf("%d",&w);  
    switch (w) {  
        case 16: printf (" Alger " );  
        break;  
        case 30: printf (" Oran " );  
        break;  
        case 31: printf (" Ouargla " );  
        break;  
        case 25: printf (" Constantine " );  
        break;  
        default: printf(" invalide code " );  
    }  
}
```